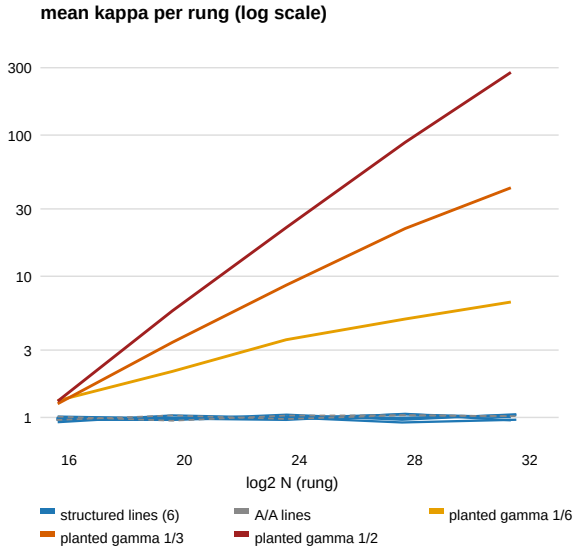
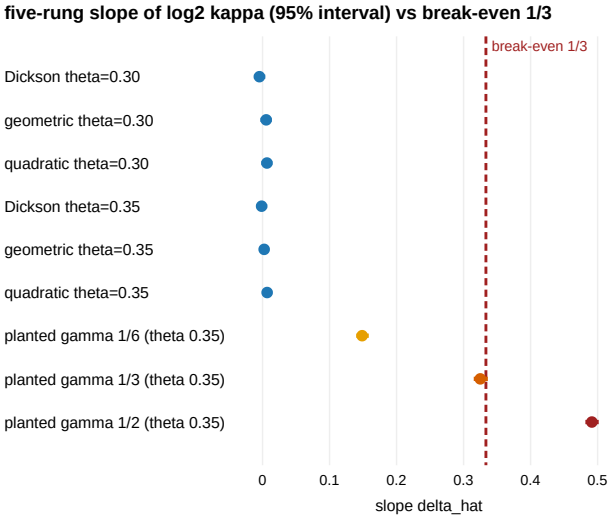
Dickson, geometric and quadratic x-bases do not show growing pair-sum coincidences either

Reading: `0-SUB-BE` for `EXP-PFDR-2d36f9-R2-FAM`. All six structured lines (3 families × θ 0.30, 0.35) are valid (smallest rung mean coincidence count 2901 ≥ 2500), the selftest and all blocking controls (C3 plants, C4 Z/N, C5 A/A, C6 z_cal) pass, no line is ALIVE and no job failed (19,887 jobs). Five-rung slopes of $\log_2\kappa$ on $\log_2 N$ are between −0.005 and +0.007, with the largest upper 95% limit 0.0147 against the break-even rate 1/3.

What was run. The same R2 ladder (kernel, curves, cells, statistics, controls, readings; `PROTOCOL.md`, frozen in commit b159698c17 before data) with the structured base replaced by three families built from the abscissa stream: Dickson $x_k = u^k + u^{-k}$, a geometric progression $x_0\rho^k$, and a quadratic progression $x_0 + k^2$. One amendment before analysis: a first census attempt aborted on a cycling Dickson/geometric stream (parameter of small multiplicative order); the fix resamples such parameters (4 of 19,887 jobs needed one resample). The partial data of the aborted attempt are kept unmodified in `aborted-attempt1/`.





Structured lines (κ per rung 16 / 20 / 24 / 28 / 32 bits; slope with 95% cluster-bootstrap interval)

line	class	slope [95%]	κ per rung
Dickson (u^k + u^-k) θ=0.30	SUB-BE	-0.0047 [-0.0130, 0.0041]	0.99 / 0.99 / 1.00 / 0.92 / 0.96
geometric progression (x0 rho^k) θ=0.30	SUB-BE	0.0054 [-0.0031, 0.0136]	0.98 / 0.97 / 1.05 / 0.96 / 1.05
quadratic progression (x0 + k^2) θ=0.30	SUB-BE	0.0066 [-0.0017, 0.0147]	0.96 / 0.96 / 1.01 / 0.99 / 1.04
Dickson (u^k + u^-k) θ=0.35	SUB-BE	-0.0014 [-0.0088, 0.0070]	1.01 / 0.98 / 0.96 / 1.06 / 0.95
geometric progression (x0 rho^k) θ=0.35	SUB-BE	0.0024 [-0.0049, 0.0098]	1.00 / 0.98 / 1.00 / 0.99 / 1.03
quadratic progression (x0 + k^2) θ=0.35	SUB-BE	0.0068 [-0.0004, 0.0144]	0.92 / 1.03 / 0.99 / 1.05 / 1.01

Controls

line	class	slope [95%]	κ per rung
gamma = 0 θ=0.30	SUB-BE	-0.0046 [-0.0115, 0.0028]	1.27 / 1.42 / 1.33 / 1.27 / 1.27

gamma = 1/6 θ=0.30	SUB-BE-GROWING	0.1489 [0.1434, 0.1545]	1.30 / 2.45 / 3.46 / 4.85 / 7.29
gamma = 1/3 θ=0.30	SUB-BE-GROWING	0.3137 [0.3087, 0.3193]	1.29 / 4.05 / 8.52 / 19.56 / 45.21
gamma = 1/2 θ=0.30	ALIVE	0.4859 [0.4811, 0.4911]	1.25 / 6.71 / 21.66 / 80.60 / 300.80
gamma = 0 θ=0.35	SUB-BE	-0.0077 [-0.0149, -0.0001]	1.31 / 1.30 / 1.32 / 1.21 / 1.22
gamma = 1/6 θ=0.35	SUB-BE-GROWING	0.1486 [0.1399, 0.1585]	1.32 / 2.12 / 3.55 / 4.98 / 6.57
gamma = 1/3 θ=0.35	ALIVE	0.3249 [0.3152, 0.3362]	1.25 / 3.40 / 8.65 / 21.81 / 42.37
gamma = 1/2 θ=0.35	ALIVE	0.4914 [0.4818, 0.5016]	1.30 / 5.69 / 22.08 / 88.80 / 278.17
Z/N interval θ=0.30	ALIVE	0.6986 [0.6961, 0.7012]	1150.27 / 7888.69 / 55768.53 / 381836.44 / 2626102.77
Z/N interval θ=0.35	ALIVE	0.6444 [0.6362, 0.6539]	680.24 / 3995.91 / 23375.65 / 155756.82 / 748921.44
Dickson (u^k + u^-k) θ=0.30	SUB-BE	0.0048 [-0.0033, 0.0134]	0.96 / 1.03 / 0.97 / 1.04 / 1.02
geometric progression (x0 rho^k) θ=0.30	SUB-BE	0.0048 [-0.0036, 0.0134]	0.96 / 1.03 / 0.97 / 1.04 / 1.02
quadratic progression (x0 + k^2) θ=0.30	SUB-BE	0.0048 [-0.0035, 0.0137]	0.96 / 1.03 / 0.97 / 1.04 / 1.02
Dickson (u^k + u^-k) θ=0.35	SUB-BE	0.0039 [-0.0043, 0.0121]	1.01 / 0.95 / 1.03 / 1.04 / 1.02
geometric progression (x0 rho^k) θ=0.35	SUB-BE	0.0039 [-0.0039, 0.0120]	1.01 / 0.95 / 1.03 / 1.04 / 1.02
quadratic progression (x0 + k^2) θ=0.35	SUB-BE	0.0039 [-0.0042, 0.0123]	1.01 / 0.95 / 1.03 / 1.04 / 1.02

The instrument sees growth (planted γ = 1/6, 1/3, 1/2 recovered as 0.149, 0.314–0.325, 0.486–0.491; Z/N 0.699 and 0.644) and reads γ = 0 and all A/A lines as SUB-BE. **C6:** n = 10, mean 0.219, sd 0.879 (bounds |mean| ≤ 0.62, sd in [0.548, 1.454]); the original 30-cell rule gives mean 0.219, sd 0.834.

Scope and what it says

- With the window / rank-d label-GAP lines of the rerun (PR #3188) this makes twelve structured x-base lines (six families × two θ) over 16–32 bits with valid controls, none showing κ growth with N: the pair-sum coincidence excess of these bases tracks a random base. Obstruction: max upper95(δ) = 0.0147 against 1/3.
- **Not covered, and why.** The FAM-1 mitm-engine arms (`subgroup` , `small_x` , `dickson` of EXP-PFDR-011cd0) are not reproduced: that engine is not in this repository, the `small_x` arm is the x-window line already measured, and `subgroup` needs a proper subgroup of the curve group (these curves have prime order); the Dickson line here is a base-construction analogue, not that arm. The bbc53e-base is the planted shared-unknown base: it is the γ control (recovered slopes above), a plant for κ growth by construction, not a candidate family.
- Sizes above 32 bits and other families are not covered. It prices no end-to-end index calculus and says nothing about rho. Run as a study in this repository (local, counts deterministic); the lead disposition is the sibling repository's call. Raw jobs are in the artifact store with manifests.